

Diversity Gain of 2-D PD Array in a 13.3°/s High-mobility FSO System with 13-ms Handover Latency

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Abstract We investigate the impact of 2-D PD array-based diversity reception in realistic mobile FSO scenarios, including terminal mobility and handovers. By eliminating tracking jitter-induced outages through diversity combining, the arrayed receiver achieved zero tracking-induced interruption under a 13.3°/s terminal velocity and 13-ms handover latency. ©2026 The Author(s)

Introduction

Leveraging advanced beam-steering and tracking technologies, free-space optical (FSO) communication offers unprecedented spatial density, capacity, and security for 6G hotspots and data-center applications [1-6]. Crucial to these mobile and switchable FSO systems are pointing, acquisition, and tracking (PAT) mechanisms [5,6], which maintain link alignment despite transceiver sway and platform vibrations. However, as data rates approach 100 Gbps, even 10- μ m-order beam-spot fluctuations—inherent in the residual jitter of PAT processes—become critical. Such deviations are significant relative to the tens-of-microns apertures of high-speed photodetectors (PDs) and fiber cores, resulting in severe power penalties. Furthermore, high-speed adaptive optics often deform the received beam profile, further complicating PAT. Despite their impact on link stability, such ‘micro’ interruptions have often been tolerated in prior works. In practice, suppressing such kHz-class micro-fluctuations solely through PAT hardware improvement is cost-prohibitive, necessitating a holistic design that

synergizes PAT with spatial-diversity receivers.

So far, we have proposed and demonstrated a spatial diversity receiver using a high-speed two-dimensional PD array (2-D PDA) [7-9] for overcoming the micro fluctuations in mobile FSO systems. This study further evaluates diversity gains in complex mobile scenarios, specifically focusing on terminal switching, or handover. By integrating a standard beacon-assisted fast steering mirror (FSM)-based PAT system with the 2-D PDA receiver, we achieved zero interruption time (excluding hard handover losses) under rigorous conditions: a terminal translation speed of 13.3°/s and a handover latency of around 13 ms.

Experimental setup

Fig. 1 shows the emulation platform to assess the robustness of the free-space optical link under terminal translation and handover. The setup consists of two transmitters: one remains stationary (TX1), while the other is mounted on a linear stage to facilitate lateral motion from Y = +300 to -700 mm at 100 to 400 mm/s (TX2), and one fixed receiver (RX). The distances from RX to TX1 and

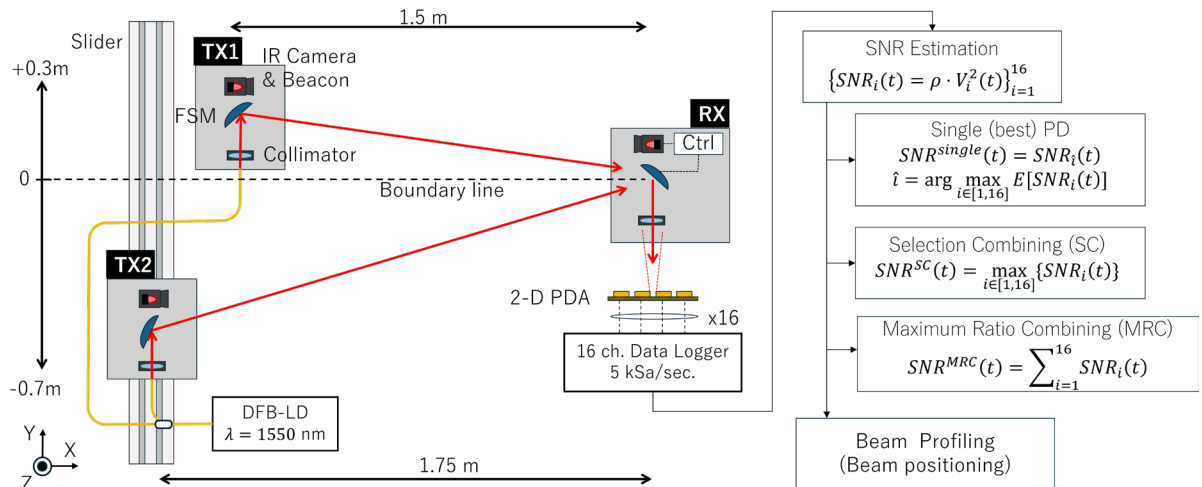


Fig. 1 Experimental setup. TX1 was stationary while TX2 was mounted on a linear stage. RX dynamically selected the active transmitter depending on the position of TX2. FSM: fast steering mirror, 2-D PDA: two-dimensional photodetector array.

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TX2 were set at 1.50 m and 1.75 m (at $Y = 0$), respectively. The output from a 1550-nm distributed feedback laser diode (DFB-LD) was split between TXs, where it was collimated ($\phi = 8$ mm) and emitted toward RX.

To enable high-speed beam steering and tracking, each transceiver terminal was equipped with an FSM driven by a positioning system comprising a beacon light and an IR camera (1936×1216 , FoV = $\pm 36^\circ$). The cycle time of the control loop was around 10 ms. Depending on the real-time position of TX2, RX dynamically selected the active transmitter: employing TX2 for $Y < 0$ and initiating a handover to TX1 as TX2 crosses the $Y = 0$ boundary, namely the cell edge.

The received light was de-collimated and detected by a 2-D PDA consisting of a 4×4 square-grid of $30\text{-}\mu\text{m}$ square pixels with a $44\text{-}\mu\text{m}$ pitch [7]. The 3-dB bandwidth and responsivity of the PD elements was 15 GHz and 0.35 A/W, respectively. The received optical power from each TX was +6 dBm and the beam spot radius at the PDA was around $80\text{ }\mu\text{m}$. However, the actual coupling efficiency was contingent upon beam profile and pixelation error as discussed in [5]. After passing through shunt resistors, the 16 ($=4 \times 4$) outputs from the PDA were simultaneously sampled at 5 kHz and stored via a 16-ch data logger (Keyence

NR-X with ST-04 modules).

Note that, as high-speed oscilloscope memory is insufficient to capture dynamics over several seconds, an indirect measurement approach without data modulation was adopted. In addition, the optical paths were assumed to be symmetric. We estimated the transmission performance by mapping the received powers to the actual SNR based on our 40-Gbps transmission experiment [5,6], i.e., $\{SNR_i(t) \cong \rho \cdot V_i^2(t)\}_{i=1}^{16}$, where $V_i(t)$ denotes the output voltage from the i -th PD element at time t and $\rho (= 5.23 \times 10^{-5})$ is the empirical coefficient for the SNR estimation. Three diversity reception schemes were considered:

- **Single (best) PD:** only the single PD element with the highest average received power was utilized. This scheme serves as a baseline, corresponding to a conventional FSO system without spatial-diversity receivers,
- **Selection Combining (SC):** a low-complexity diversity receiver solution where the single PD element with the highest instantaneous received power is selected adaptively, and
- **Maximum Ratio Combining (MRC):** This scheme optimally weights and sums the signals from all 16 PD elements to maximize the combined signal-to-noise ratio (SNR).

The SNR estimation procedures for each diversity reception scheme are illustrated in Fig. 1.

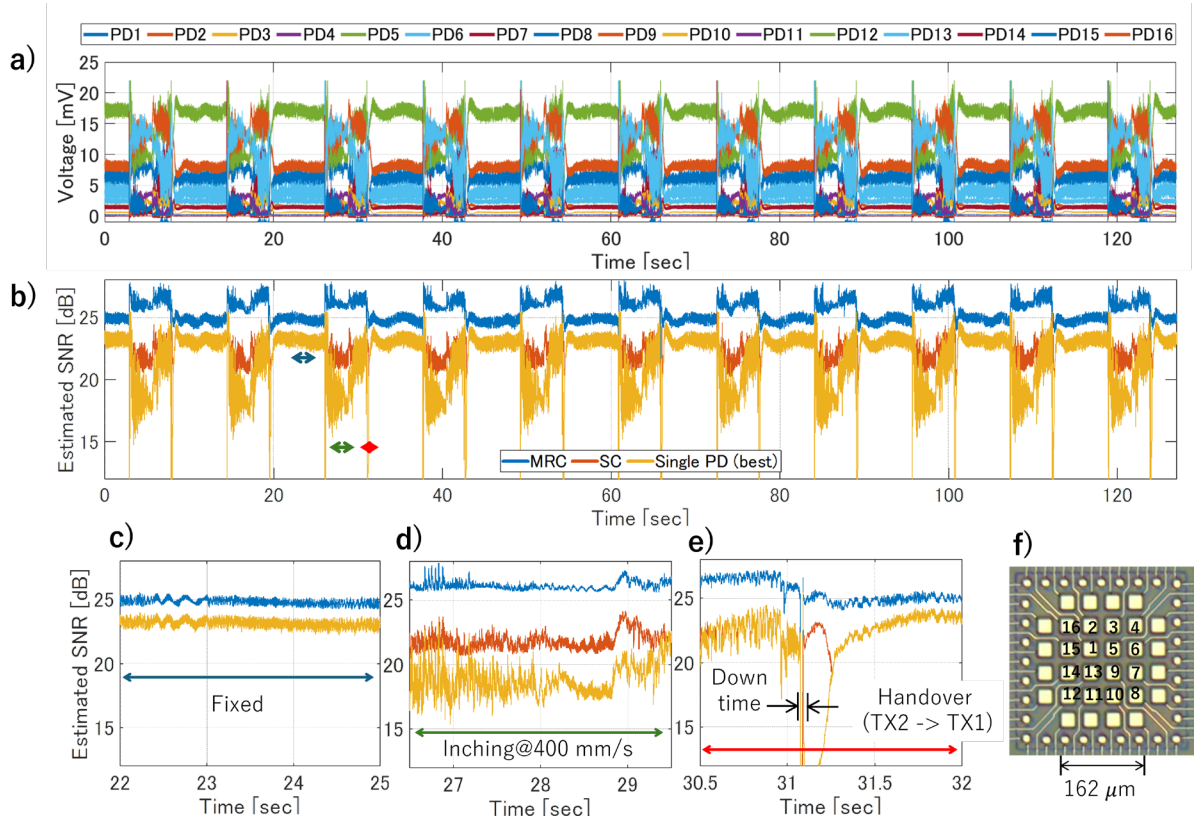


Fig. 2 Temporal evolution of a) the output voltage per PD and b) the estimated SNR after diversity combining while TX2 is translated at 400 mm/s. Magnified views of the SNR curves for a) fixed (TX1), b) inching (TX2), and c) switching (TX2 $\rightarrow$ TX1). f) Photograph of the 2-D PDA device (16 were used out of the 32 available elements).

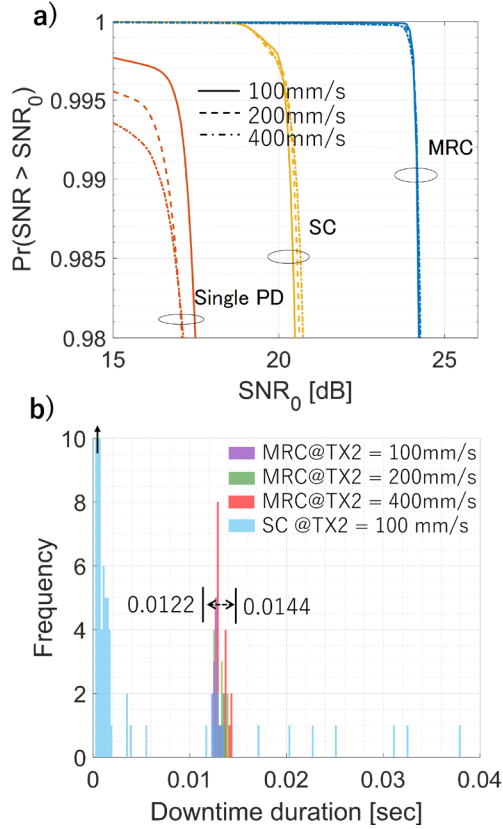


Fig. 3 a) CCDF of the estimated SNR and b) histogram of the downtime duration for SC and MRC.

Experimental Results

Fig 2. illustrates the temporal evolution of a) the output voltage per PD and b) the estimated SNR with the three different diversity combining methods, recorded while TX2 was translated at 400 mm/s. When TX1, the fixed terminal, was activated, the SNR fluctuations remained minimal (Fig. 2c), with negligible differences observed among the three schemes. The beacon-camera-based PAT was enough for accurately coupling the transmitted beam into a single 30- μm square PD element in this case. However, when TX2, the mobile terminal, was selected, maintaining a stable coupling to the single PD element (yellow) became challenging, causing the significant SNR fluctuations (Fig. 2d). Furthermore, during the handover, the inability to follow the FSM's transient dynamics leads to a prolonged downtime (Fig. 2e). The SNR fluctuations were significantly suppressed through SC (red). Nevertheless, SC failed to compensate for power loss when the beam resides in the inter-pixel gaps or undergoes deformation, leading to frequent microsecond-scale link losses, particularly during terminal movement. (Fig. 2d). MRC (blue) maximized the spatial diversity gain of the PDA. Beyond simply boosting the effective received power, it provided robust stabilization of the link characteristics, ensuring high performance regardless of the terminal motion or beam profile variations.

Statistically, as shown in the complementary cumulative distribution function (CCDF) of Fig. 3a, the use of MRC achieved an SNR of 20 dB or higher with a probability of 99.98%, even when TX2 was translated at its maximum stage velocity of 400 mm/s. This speed is equivalent to an angular velocity of 13.3°/s. Fig. 3b is the histogram of the duration of the downtime ($\text{SNR} < 20$ dB). With MRC, all observed downtime were confined around 13 ms, which was consistent with the FSM control cycle. This indicates that the 0.002% downtime for MRC was entirely attributable to unavoidable disconnections during hard handovers; the residual jitter inherent in PAT processes is completely compensated for.

In contrast, SC suffered from frequent millisecond-scale downtimes even at a TX2 translation speed of 100 mm/s as in Fig. 3b. Fig. 4 shows the power spectrum of the beam centroid trajectory, estimated using the 2-D PDA as an image sensor. One can find significant spectral peaks between 0.3 and 0.7 kHz. These peaks were attributed to the intrinsic oscillations of the FSM during high-speed tracking control rather than the vibration induced by the terminal's movement itself. Such fluctuations induce millisecond-scale downtimes in SC (and Single PD) as in Fig. 3b. These micro interruptions were entirely offset as MRC effectively leverages the full diversity gain of the PDA. This highlights a cost-effective alternative to hardware-heavy PAT enhancements, providing a viable path toward robust FSO connectivity in real-world mobile environments.

Conclusions

This study investigated the impact of the 2-D PDA based spatial-diversity reception in realistic mobile FSO scenarios, including terminal mobility and handovers. By leveraging a typical beacon-camera tracking setup alongside the 2-D PDA based receiver with MRC, we achieved stable connectivity at terminal speeds up to 13.3°/s, where no outages occurred due to alignment errors, except for the unavoidable 13-ms interruption during hard handovers.

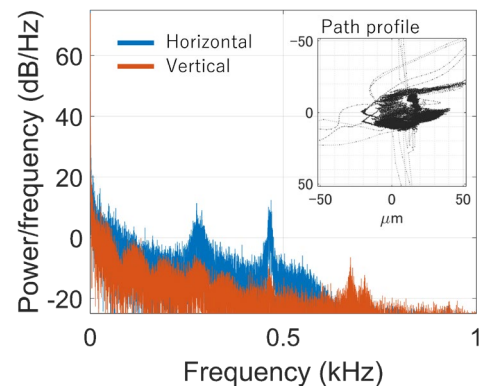


Fig. 4 Periodogram of the beam centroid displacement under TX2 translation at 400-mm/s. The inset displays a 20-second path profile of the centroid on the 2-D PDA.

Acknowledgements

This research was partially supported by the Japan Science and Technology Agency (JST) CRONOS (grant number: JPMJCS24N4, Japan), and by the Innovative Science and Technology Initiative for Security, ATLA (grant number: JPJ004596, Japan).

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